

Core specialisms

ISO, quality & technical

- ISO 15189, 17025 & 17043 (train, assess, implement)
- QMS governance, documentation & audit readiness
- Technical assessment of laboratories
- Evidence validation & data integrity

Learning, curriculum & pedagogy

- Instructional design & blended learning
- Apprenticeship & RQF curriculum (L3–L6)
- Competence frameworks & assessment strategy
- Metacognitive & learning-science models

Education

MSc Biomedical Sciences
(Immunohaematology) — Merit
Cardiff Metropolitan University · 2016

BSc (Hons) Biomedical Sciences — First Class
Cardiff Metropolitan University · 2007

BSc Biology — 2:1
University of Portsmouth · 2003

Registration

- HCPC Registered Biomedical Scientist
- Chartered Scientist (CSci)
- IBMS Full Member

Professional profile

Biomedical scientist and clinical education specialist with a strong diagnostic background across haematology, transfusion, coagulation and haemoglobinopathy screening. Over a decade in ISO 15189 and CPA-accredited NHS laboratories, progressing into technical assessment, quality systems and specialist training. Now founder of Forged Frameworks, translating scientific and regulatory rigour into accessible, high-quality learning.

70+	650+	240+	40%
ISO 15189 & 17025 lab assessments	Training days delivered per year	Hours of validated curriculum, 8 modules	Content production time cut via AI automation

Signature projects

- ISLH

Point-of-care analyser evaluation — Comparative study of the Yumizen H500 FBC analyser for paediatric point-of-care testing, presented at ISLH.
- £42k/yr

All-Wales screening algorithm — Redesigned sickle cell & thalassaemia antenatal screening — ~6 hours saved per patient contact.
- ISO 15189

CPA → ISO 15189 transition — Led change control, method validation, measurement uncertainty, budgets and quality documentation to achieve laboratory accreditation.

Professional experience

click a role to expand

- Biomedical, Diagnostic & Managerial Roles

Feb 2004 – Jul 2018

NHS Wales — Cardiff & Vale UHB

14 years in NHS pathology — from Medical Laboratory Assistant to acting senior and manager roles across haematology, transfusion and haemoglobinopathy.
- Technical Assessor & Quality Specialist

Nov 2017 – 2024

UKAS

70+ ISO 15189/17025 lab assessments across healthcare, PT, forensics & calibration; UK and international.

Further specialisms

Leadership & collaboration

- Project leadership & cross-functional coordination
- Stakeholder engagement — employers, SMEs, regulators
- Coaching, mentoring & capability development
- Commercial training delivery & consultancy

AI & digital learning

- Agentic AI authoring & content pipelines
- Bespoke HTML/CSS/JS blocks & xAPI wrappers
- Rise 360, Storyline, SCORM, xAPI, Moodle
- Accessibility (WCAG 2.2, EN 301 549)

Certifications

- TAP Diploma in Learning Design (2021)
- IBMS Certificate of Expert Practice in Training (2017)
- IBMS Specialist Diploma in Haematology & Transfusion (2010)
- UKAS Technical & Lead Assessor — ISO 15189, 22870, 17043

Platforms & tools

Laboratory & diagnostics

Automated haematology · coagulation · transfusion · haemoglobinopathy screening · EQA / IQA

Quality systems

Q Pulse · iPassport · Telepath / Dedalus · Ideagen

Authoring & media

Articulate 360 (Rise, Storyline) · SCORM · xAPI · Visme · Canva · Vyond · Descript · Camtasia

Code & AI

Python · HTML/CSS/JS · Claude Code · OpenAI Codex · OpenAI & Gemini APIs · Google Apps Script · NotebookLM

Values & soft skills

Collaborative

Analytical

Creative problem-solver

Reflective practitioner

Inclusive communicator

Continuous learner

Integrity-driven

Career history

click a role to expand

Assessment Manager, Medical Laboratories

July 2018 – Feb 2020

UKAS (Healthcare)

Planned and led ISO 15189 assessments for clinical and diagnostic laboratories.

Senior Training Manager

Feb 2022 – Sept 2024

UKAS Training Academy

650+ training days/yr on ISO 15189, 17025 & 17043; led trainers, budgets and digital transformation.

Founder & Creator

Aug 2025 – Present

Forged Frameworks

AI-driven learning-design consultancy — agentic pipelines, bespoke HTML blocks and accessibility consultancy.

Learning Content Manager

Sept 2024 – Aug 2025

Tiro Training Ltd

240+ hours of L3–L6 blended learning across six technical sciences; Moodle redesign and GenAI-assisted authoring.

Training Manager

Feb 2020 – Feb 2022

UKAS Training Academy

Migrated the UKAS Academy portfolio to virtual delivery; built healthcare training content.

Founder & Blacksmith

2020 – Present

Wig's Wonky Forge

My own creative-metalwork and blacksmithing brand — hand-forged ironwork and a growing craft side-business.

Continuing professional development

GenAI in Learning Design (Coursera / Google) · Accessibility & Universal Design for Learning · ISO 15189:2022 Implementation & Internal Audit · Prompt Engineering for L&D · AI Literacy for Educators.

Interests

Sci-fi, science and literature. Creative metalwork and blacksmithing — building my own forge and metalwork brand. Walking, running and wellbeing. Mental-health advocacy, and two very opinionated cats.

